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GUIDELINES - EXPERT OPINION

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Large B-cell lymphoma (LBCL): EHA Clinical Practice Guidelines for diagnosis, treatment, and follow-up

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Abstract

Large B-cell lymphoma (LBCL) accounts for about one-third of adult lymphoma cases. Diagnosis requires specialized hematopathology laboratories, with immunophenotypic analysis essential for confirming B-cell lineage and identifying variants. *MYC* and *BCL2* rearrangements indicate a poor prognosis. Staging and prognosis rely on positron emission tomography computed tomography (PET-CT). The International Prognostic Index (IPI) aids risk stratification. PET-CT is critical for assessing treatment response and guiding strategies. First-line management for LBCL can be informed by interim PET to assess chemosensitivity, with rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP) or polatuzumab vedotin rituximab, cyclophosphamide, doxorubicin, and prednisone (Pola-R-CHP) for advanced stages depending on IPI scores. Primary mediastinal B-cell lymphoma (PMBCL) management favors R-CHOP given every 14 days (R-CHOP14) or dose-adjusted etoposide, doxorubicin, vincristine, cyclophosphamide, prednisone, and rituximab (DA-EPOCH-R) without radiotherapy in complete responders. Elderly patients, unfit or not (≥80 years or <80 with poor fitness), need geriatric assessment to guide therapy, often R-miniCHOP or non-anthracycline regimens. Frail patients should have adapted treatments. Prephase corticosteroids improve performance status, and supportive treatment should be optimized. The value of central nervous system (CNS) prophylaxis remains uncertain. CNS-IPI scores and specific anatomical sites help identify high-risk patients; magnetic resonance imaging (MRI) and cerebrospinal fluid (CSF) analysis are recommended. Approximately 30%–40% of patients with LBCL experience relapsed or refractory disease after 1L treatment. Treatment strategies vary based on the timing of relapse (<1 year or ≥1 year). For those refractory or relapsing within <1 year and fit for therapy, chimeric antigen receptor T (CART) are the gold standard in 2L. CART in CART-naïve patients and bispecific antibodies appear to be the best approach in 3L. Follow-up includes clinical examination for 2 years and management for long-term side effects, such as cardiotoxicity, osteoporosis, immune dysfunction, neurocognitive impairment, endocrine dysfunction, fatigue, neuropathy, and mental distress.

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METHODOLOGY

These guidelines aim to provide evidence-based recommendations for the diagnosis, treatment, follow-up, and supportive care of patients with large B-cell lymphoma (LBCL) in Europe. They are intended for healthcare professionals. A multidisciplinary panel of experts was formed, including hematologists, pathologists, biologists, radiotherapists, nuclear medicine physicians, and patient representatives. The team ensured geographic and professional diversity to reflect variations in healthcare systems across Europe. Roles were assigned for drafting, evidence review, and editorial oversight.

A systematic review of the literature was conducted using PubMed. The review prioritized high-quality evidence. Evidence was graded using the GRADE framework to assess the strength and quality of recommendations (see Appendix Table A1). Draft recommendations were developed based on evidence and circulated among the panel for feedback. Discrepancies were resolved through a series of virtual meetings and consensus discussions. These guidelines emphasize patient-centered care, integrating shared decision-making, patient-reported outcomes, and psychosocial support to enhance experiences and align care with patient needs. These guidelines will be updated every 2 years and treatment algorithms annually on the EHA website.

INCIDENCE AND EPIDEMIOLOGY

LBCL, including mostly diffuse large B-cell lymphoma (DLBCL), accounts for approximately one-third of all lymphomas in adults.[1] In Europe, the incidence rate for LBCL is 28,000/year,[2] with more males being affected compared to females.[1] The incidence increases with age. In the vast majority of patients, the etiology of LBCL remains unclear. Various risk factors for the development of LBCL such as immunodeficiency, B-cell activating autoimmune diseases, virus infections, or a family history of lymphoma have been identified[3] (Table 1).

DIAGNOSIS, PATHOLOGY, AND MOLECULAR BIOLOGY

LBCLs, including DLBCLs not otherwise specified (NOS), should be diagnosed in a reference hematopathology laboratory with expertise in the diagnostic work-up of lymphoma specimens (Table 2). In particular, the full spectrum of immunohistochemical markers and molecular techniques should be available to render a lymphoma diagnosis according to the current World Health Organization (WHO)[6]/International Consensus Classification (ICC).[4]

Essential diagnostic criteria comprise a characteristic morphology and infiltration pattern together with evidence of a mature B-cell phenotype by immunohistochemistry (IHC) or flow cytometry (or both). These criteria support the diagnosis of LBCL, its various subtypes, and, eventually, transformation from indolent lymphomas. The IHC panel should encompass markers that allow distinction from alternative diagnoses, such as Burkitt lymphoma, plasmablastic lymphoma, blastic/pleomorphic mantle cell lymphoma, and rare variants of Hodgkin lymphoma. In addition, more specific subgroups of LBCLs should be identified (e.g., EBV-positive DLBCL, primary mediastinal LBCL). A statement on the cell of origin (COO) classification in the diagnostic report (e.g., by an immunohistochemical classifier) is considered desirable by the WHO and ICC classifications. However, IHC misclassifies up to 30% of patients. Therefore, restricting access to potentially active therapeutic options based solely on IHC is not recommended.

It is highly recommended to use genetic/molecular testing to separate DLBCL, NOS from the diagnostic category of LBCL with MYC and BCL2 and/or BCL6 rearrangements, by fluorescence in situ hybridization (FISH) for the MYC locus, and, if positive, by subsequent FISH testing for BCL2 and BCL6 rearrangements. Both recently revised classifications[4,5] describe HGBCL with MYC and BCL2 rearrangements (with or without BCL6 rearrangement) as an aggressive lymphoma of GCB origin with distinct biology from other LBCLs. Evidence supporting a distinct biology in patients with MYC and BCL6 rearrangements is currently limited. In the WHO-HAEM5,[5] dual MYC and BCL6 rearrangements are now classified either as a subtype of DLBCL, NOS or HGBL, NOS according to their cytomorphological features. The revised ICC[4] has retained HGBCL-DH-BCL6 as a provisional entity justifying further studies. MYC translocation to an immunoglobulin partner is most strongly associated with inferior overall survival (OS).[7,8] Other predictors of poor outcome such as TP53 mutations and "molecular high-grade" gene expression signature are not yet routinely used in clinical prognostication. It is noteworthy that more specific subtypes, for example, LBCLs with IRF4 rearrangement of high-grade B-cell lymphoma with 11q aberration, reflect the heterogeneity of LBCL, but currently have no therapeutic implications.

The two molecular subtypes[9] including activated B-cell-like DLBCL (ABC DLBCL) and germinal center B-cell-like DLBCL (GCB DLBCL)[10] associated with different COO may be determined by IHC at diagnosis, even if various initial trials that investigated the addition of specific drugs to front-line rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP) therapy failed to show a positive signal,[11] contrary to more recent approaches[12,13] or at the time of relapse.[14] Recently, novel molecular subtypes were identified using next-generation sequencing approaches.[15,16] However, the identification of these subtypes is not applicable at this stage in clinical routine and their prognostic and potentially predictive value needs to be established in prospective clinical trials. These double expressors are associated with poor outcomes in patients treated with R-CHOP[17]; however, no specific treatment is currently recommended, as they do not represent distinct biological subgroups but rather the convergent result of diverse underlying pathways.[4]

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TABLE 1 Reported risk factors for large B-cell lymphoma (adapted from ref. 3).

Risk factors	Odds ratio (OR)	95% CI
Inherited immunodeficiency syndrome, HIV.AIDS, organ transplant recipients	>2	NA
B-cell activating autoimmune diseases (e.g., Sjögren syndrome, systemic lupus erythematosus, and rheumatoid arthritis)	2.36	1.80–3.09
Viral infections (e.g., HIV, KSHV/HHV8, HCV, EBV, and HBV)	2.02	1.47–2.76
Family history of non-Hodgkin lymphoma	1.95	1.54–2.47
Obesity: Young adult BMI (30+ vs. 18.5–22.4 kg/m^2)	1.58	1.12–2.23
Recreational sun exposure (inverse association)	0.78	0.69–0.89
Any atopic disorder (inverse association)	0.82	0.76–0.89
Higher socioeconomic status (inverse association)	0.86	0.79–0.94
Multiple environmental exposures, for example, trichloroethylene, benzene, and pesticides and herbicides, glyphosate, field crop/vegetable farm worker (women)	1.78	1.22–2.60
Hairdresser (women)	1.65	1.12–2.41
Seamstress/embroiderer (women)	1.49	1.13–1.97
Low adult BMI (<18.5 vs. 18.5–22.4 kg/m^2, women)	0.46	0.29–0.74
Hormone replacement therapy (50 years, women)	0.68	0.52–0.88
Oral contraceptive use before 1970 (women)	0.78	0.62–1.00
Material handling equipment operator (men)	1.58	1.02–2.44
Lifetime alcohol consumption (>400 kg vs. nondrinker, men)	0.57	0.44–0.75
Previous blood transfusion (men)	0.69	0.57–0.83

Abbreviations: BMI, body mass index; EBV, Epstein-Barr virus; HBV, Hepatitis B virus; HCV, Hepatitis C virus; KSHV/HHB8, Kaposi's Sarcoma-associated Herpes Virus/Human Herpes virus type 8; NA, not applicable.

STAGING, PROGNOSTIC/RISK ASSESSMENT

Positron emission tomography computed tomography (PET-CT)[18,19] is the primary imaging method for staging, efficiently identifying extranodal disease-impacting prognosis,[20] except in leptomeningeal involvement or parenchymal brain, where magnetic resonance imaging (MRI) is the gold standard. For patients at high risk of CNS relapse, baseline MRI and CSF analysis are recommended to exclude occult secondary CNS involvement.

PET-CT surpasses bone marrow biopsy in detecting bone marrow involvement with LBCL and is highly specific.[21,22] The presence of non-fluorodeoxyglucose (FDG)-avid bone marrow disease does not confer a worse prognosis.[23–26] Bone marrow biopsy may be considered for selected patients in whom a coexisting hematological condition is suspected (e.g., low-grade lymphoma or myelodysplasia), and where this would inform clinical management, but is otherwise unnecessary.

Multiple prognostic indexes derived from the International Prognostic Index (IPI) have been reported.[27–29] However, IPI is mandatory and CNS-IPI is strongly recommended before treatment[30,31] (Table 3). PET-CT provides a precise evaluation of baseline tumor burden assessed by the standardized uptake value (SUV) 4.0 threshold-based metabolic tumor volume (MTV), a highly promising prognostic biomarker in LBCL,[32–35] as well as ctDNA.[36] New indexes including MTV have been developed that better discriminate patients over IPI.[37–40] Quantitative measures such as maximum lesion distance (Dmax) and distance between the largest and another lesion (Dmaxbulk)[41,42] offer potential for enhanced risk stratification.

ADDITIONAL BASELINE INVESTIGATIONS AND MANAGEMENT

Enhancing all aspects of supportive care is crucial to minimize morbidity, especially in elderly or frail patients (Table 4). Patients often present with complex needs, and involvement of key members of the wider multidisciplinary team (MDT) (e.g., lymphoma clinical nurse specialists, pharmacists, cardiorespiratory specialists, and healthcare of older people liaison teams) can prove invaluable.

The risk of osteoporotic bone fractures is significant in the LBCL patient population.[43] A predisposing history (osteoporosis, osteopenia, prior fracture, and rheumatoid arthritis), bone involvement with lymphoma, and prephase corticosteroid therapy are independent risk factors. Baseline osteoporosis risk should be considered (e.g., the FRAX score). Patients receiving steroid therapy have been shown to benefit from vitamin D treatment in other contexts, so this may be considered.[44,45]

Simultaneous administration of calcium channel blocking agents could adversely impact the antitumor efficacy of anti-CD20-based monoclonal antibodies.[46] Alternative drugs could be considered in these patients.

Patients should receive clear, accessible information on their diagnosis and treatment options, enabling them to express their goals and preferences regarding quality of life and functional status, and to actively participate in their care. Psychosocial distress screening should be performed at diagnosis using a validated tool (e.g., NCCN Distress Thermometer, Hospital Anxiety and Depression Scale). Identifying distress early allows timely referrals to psychological and supportive care services.

RESPONSE ASSESSMENT

PET imaging, per Lugano criteria,[18,19,47] is preferred for response assessment (Table 5), with the timing varying depending on the specific LBCL entity (Figure 1). End-of-treatment (EoT) PET predicts outcomes,[48] with scores of 1–3 on the five-point scale (Deauville Score [DS]) indicating complete metabolic response (CMR). Although EoT PET scans show high negative predictive value (NPV > 90%), the suboptimal positive predictive value (~60%) warrants biopsy or follow-up PET, in case of lesions with residual positive uptake.

Interim PET scans (iPET2, iPET4) after 2–4 R-CHOP cycles may be helpful to establish chemosensitivity early, as it is associated with a strong NPV,[49] using the ΔSUVmax reduction criteria. Reports suggest that 94%–100% of patients achieving CMR at iPET maintain CMR at EoT PET, although MYC-positive DLBCL may show aberrant PET responses.[50]

Emerging evidence supports cell-free DNA as a promising biomarker for tumor burden and therapy response assessment.

MANAGEMENT OF 1L LBCL

Management of 1L LBCL is based on several prognostic factors, including the extent of disease dissemination. Localized or early-stage LBCL (Stage I/II) accounts for approximately 30% of newly diagnosed cases.[51] In localized disease, low-risk or high-risk disease has been conventionally informed by age (<60 years old, ≥60 years old) and the presence of adverse clinical factors (high lactate dehydrogenase [LDH] and poor performance status [PS]).

Bulky disease is widely defined as a tumor diameter of ≥7.5 cm of the largest lesion.[52] Recent data offer an alternative definition of

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TABLE 2 Baseline investigations for LBCL diagnosis, staging, and prognostic-risk assessment. (continued on next page)

Investigation/procedure	Recommendation level	Notes
Biopsy		
Surgical excision biopsy	**[I, B]**	Preferred method
Needle-core or endoscopic biopsy	**[II, B]**	Acceptable if surgery is impractical or high risk
Diagnosis by reference hematopathology lab	**[I, A]**	Access to full phenotypic and molecular investigations required
Expression of CD20, BCL6, CD10, IRF4/MUM1 MYC, BCL2, Ki67, and CD5 Expression of CD19 only at relapse	NA	Recommended[4,5]
Double expressor [MYC/BCL2]	NA	Recommended[4,5] Positive value for MYC > 40% of tumor cells Positive value for BLC2 > 50% of tumor cells
Molecular testing: FISH for MYC translocation and Ig partner FISH for BCL2 and BCL6 translocations if MYC rearranged	NA	Recommended[4,5]
EBER-1	NA	Optional based on the morphology or any relevant clinical situations
CD19 only at time of relapse	NA	Relevant for treatment choices
Blood tests		
Serum lactate dehydrogenase (LDH)	**[I, A]**	Baseline measurement
Albumin	**[I, A]**	
Quantitative immunoglobulins	**[I, A]**	
Hepatitis B, C, HIV serology	**[I, A]**	Full serology panel
Imaging and staging		
Baseline PET-CT	**[I, A]**	Preferred for staging
Contrast-enhanced CT (neck, chest, abdomen)	**[II, B]**	If PET-CT is not feasible
Staging bone marrow biopsy	**[II, B]**	Only where discordant or alternative bone marrow pathology would influence clinical management
Central nervous system (CNS) assessment		
MRI (brain/spine if indicated)	**[I, B]**	In high risk of CNS relapse
CSF assessment (cytology and immunophenotyping)	**[I, B]**	In high risk of CNS relapse
Cardiac assessment		
Electrocardiogram (ECG), and blood pressure measurement	**[I, B]**	Baseline for all patients before anthracyclines, intensive chemotherapy with autograft, CAR-T cells, or bispecific antibodies
Echocardiogram (older patients or risk factors)	**[II, B]**	Assess left ventricular function
Performance and prognostic scores		
ECOG performance status	**[I, A]**	Document at baseline
International Prognostic Index (IPI)	**[I, A]**	Document at baseline
Central nervous system-International Prognostic Index (CNS-IPI) score	**[I, A]**	Document at baseline
Metabolic tumor volume	NA	Recommended
International Metabolic Prognostic Index (IMPI)	NA	Recommended
Molecular and genetic testing		
Fluorescence in situ hybridization (FISH) for MYC	**[I, B]**	Baseline genetic profiling
FISH for BCL2 and BCL6 (if MYC+ identified)	**[I, B]**	
Cell of origin (COO) classification	**[II, B]**	Determined by IHC
Multidisciplinary team (MDT) discussion		
Review diagnosis and treatment plan at MDT	**[I, A]**	
Patient communication and support		
Fatigue assessment	NA	Baseline

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TABLE 2 (Continued)

Investigation/procedure	Recommendation level	Notes
Mental distress assessment	NA	Baseline
Discuss diagnosis with patient/family	**[I, B]**	Provide clear, accessible, and jargon-free information on the diagnosis and treatment options, benefits and risks, including the impact on quality of life, and check if patients understand the information provided. Patient preferences for their initial therapy should be discussed.

Abbreviations: CSF, cerebrospinal fluid; IHC, immunohistochemistry; MRI, magnetic resonance imaging; NA, not applicable.

TABLE 3 International Prognostic Index (IPI) and central nervous system (CNS)-IPI.

IPI	
Factor	
Age	≤60 or >60
Serum LDH	≤1× normal or >1
Performance status	0 or 1 versus 2–4
Stage	I or II versus III or IV
Extranodal involvement	≤1 site versus ≥2 sites
IPI categories	*All patients*
Low	0 or 1
Low intermediate	2
High intermediate	3
High	4 or 5
Age-adjusted IPI categories	*≤60 or >60*
Low	0
Low intermediate	1
High intermediate	2
High	3
CNS-IPI	
Factor	
Age	≤60 or >60
Serum LDH	≤1× normal or >1
Performance status	0 or 1 versus 2–4
Stage	I or II versus III or IV
Extranodal involvement	≤1 site versus ≥2 sites
Renal and/or adrenal involvement	Yes or no
IPI categories	*All patients*
Low	0 or 1
intermediate	2 or 3
High	4 or 5 or 6

Abbreviation: LDH, lactate dehydrogenase.

bulk, based on metabolic volume (total metabolic tumor volume [TMTV] > 220 mL) or, as included in the International Metabolic Prognostic Index (IMPI), as a continuous variable.[34,35]

Notwithstanding the clinical and biological heterogeneity of LBCL, most patients are currently treated with the same approaches, including patients with transformation from previously untreated (or unknown) indolent lymphomas.[53]

1L LBCL, Stage I/II, 0 risk factor, <80 years

These patients are IPI = 0 if <60 years old or aaIPI = 0 if 60–80 years old.

- **For patients with non-bulky disease**
  - In patients ≤60 years, abbreviated chemoimmunotherapy, with four cycles of R-CHOP, is highly effective. In the FLYER trial, two additional rituximab doses were administered.[54] The 3-year progression-free survival (PFS) was 96%.[54] With this approach, response assessment was by CT imaging.
  - For patients ≤80 years, a PET-adapted approach after two cycles is supported by randomized trial data (e.g., LYSA LNH 09-1B),[55] enabling abbreviated chemoimmunotherapy (R-CHOP for four cycles) for patients with CMR on iPET2. A non-randomized Phase 2 study (S1001) also supports this approach for those with iPET3 CMR.[56]
- **For patients with bulky disease (≥7.5 cm)**. R-CHOP for six cycles is a standard approach.

1L LBCL, 1 risk factor (Stage III/IV or Stage I/II with poor PS or high LDH), <80 years

These patients are IPI = 1 if <60 years old or aaIPI = 1 if 60–80 years old. They present with either a disseminated disease (Stage III/IV) without any other risk factor, or with a localized disease (Stage I/II) with 1 risk factor, either poor performance status (PS) or elevated LDH level.

- R-CHOP for six cycles is the standard approach **[I, A]**[57] (Table 6).
- In early-stage LBCL, radiotherapy (RT) is generally not indicated if six cycles of full dose intensity are delivered and CMR is achieved **[I, A]**.[58,59] Radiotherapy consolidation (involved site radiotherapy [ISRT] 30 Gy in 15 fractions) after immunochemotherapy is generally not indicated if full dose intensity is delivered because it offers limited survival benefit **[III, C]**.[58,59]
- In early-stage LBCL, there is no evidence to support adopting a different approach per se for patients with biological risk factors such as *MYC* rearrangement (with/without BCL2 rearrangements) **[III, B]**.[60–62]

1L LBCL, Stage III/IV, ≥2 risk factors, <80 years

These patients present with an IPI = 2–5.

- Polatuzumab vedotin rituximab, cyclophosphamide, doxorubicin, and prednisone (Pola-R-CHP) demonstrated a 6.5% improvement in 2-year PFS compared to R-CHOP in the POLARIX trial, making it the standard for IPI 2–5 disease in many regions **[I, B]**.[12] Pola-R-CHP for six cycles is preferred, although R-CHOP is an acceptable

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TABLE 4 Additional investigations prior to treatment.

Investigation/procedure	Recommendation level	Notes
Consider referral to relevant specialties	**[I, B]**	(Cardiology, health care of older people, endocrinology) for medical optimization prior to and/or during treatment
Osteoporosis		Consider osteoporosis risk in all patients FRAX score
Vitamin D supplementation		According to risk profile
Calcium channel blocking agents	**[III, A]**	Alternative drugs could be considered in patients treated with the anti-CD20 monoclonal antibody
Primary G-CSF prophylaxis	**[I, A]**	To all patients in curative intent, particularly patients older than 60 when treated with R-CHOP and more intensive immunochemotherapy
Primary prophylaxis against HSV/VZV	**[II, B]**	In selected patients
Primary prophylaxis against Pneumocystis jirovecii	**[II, B]**	In selected patients
Vaccination against influenza, COVID, pneumococcus, herpes zoster	**[II, B]**	Before treatment when feasible

Abbreviations: G-CSF, granulocyte colony-stimulating factor; HSV/VZV, Herpes Simplex virus/Varicella Zoster virus; R-CHOP, rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone.

TABLE 5 Response assessment.

Assessment/procedure	Recommendation level	Notes
Imaging		
FDG-PET (end of treatment [EOT])	**[III, A]**	Use the Lugano criteria for response assessment
Deauville Score (DS 1–3 = complete metabolic response, CMR)	**[III, A]**	CMR indicates favorable outcome
Incomplete metabolic response (DS 4–5)	**[III, A]**	Biopsy or repeat PET-CT at 8–12 weeks
Omit EOT PET (optional)	**[III, B]**	For patients with MYC-positive lymphomas achieving early CMR, caution is warranted as PET imaging may yield aberrant or misleading results
PET-CT (Interim, iPET2 or iPET4)	**[III, A]**	Selects patients for different therapy
ΔSUVmax > 70% at iPET4	**[III, A]**	Indicates favorable response
DS 5 (residual uptake at least 2 times the liver SUVmax) at iPET4	**[III, A]**	Consider for salvage therapy after biopsy
Additional assessments		
Biopsy (residual uptake on PET)	**[III, A]**	Confirms residual disease or false-positive
MRI (CNS involvement suspected)	**[III, B]**	Include spine imaging if indicated
Biomarkers		
Cell-free DNA (cfDNA)	**[IV, A]**	Investigational; prognostic but not clinical standard
Multidisciplinary team (MDT) review		
Discuss response at MDT	**[III, A]**	Evaluate results and plan subsequent management

Abbreviations: CNS, central nervous system; FDG-PET, fluorodeoxyglucose-positron emission tomography; MRI, magnetic resonance imaging; PET-CT, positron emission tomography computed tomography; SUV, standardized uptake value.

alternative when Pola-R-CHP is not available. Dose-dense R-CHOP (14 days) is not superior to R-CHOP given every 21 days (R-CHOP21). High-dose chemotherapy (HD-CT) with autologous stem cell transplant (ASCT) shows no survival benefit as 1L therapy.[63,64] R-CHOP + etoposide every 14 days (R-CHOEP-14) results in a 3–5-year PFS rate of >80% in Phase II trials **[II, B]**.[65,66] A recent report suggests that Pola-R-CHP is at least as effective as R-CHOEP and potentially less toxic.[67]

- In Stage I/II disease and IPI ≥2, Pola-R-CHP may be considered, recognizing that only 11% of patients with these characteristics were included in POLARIX, limiting interpretation of the early-stage subgroup.[12]

- **RT** in advanced LBCL is reserved for patients with PET-positive, single-site residual disease post-systemic therapy. PET-negative patients do not routinely require RT **[II, B]**.[68]

Additional recommendations

Infection is a common cause of morbidity and mortality in the context of LBCL therapy.[69,70] Primary granulocyte colony-stimulating factor (G-CSF) prophylaxis and antimicrobial prophylaxis should be considered for all patients receiving chemo-immunotherapy with curative intent, particularly patients older than 60 when treated with R-CHOP and more intensive immunochemotherapy regimens.[71]

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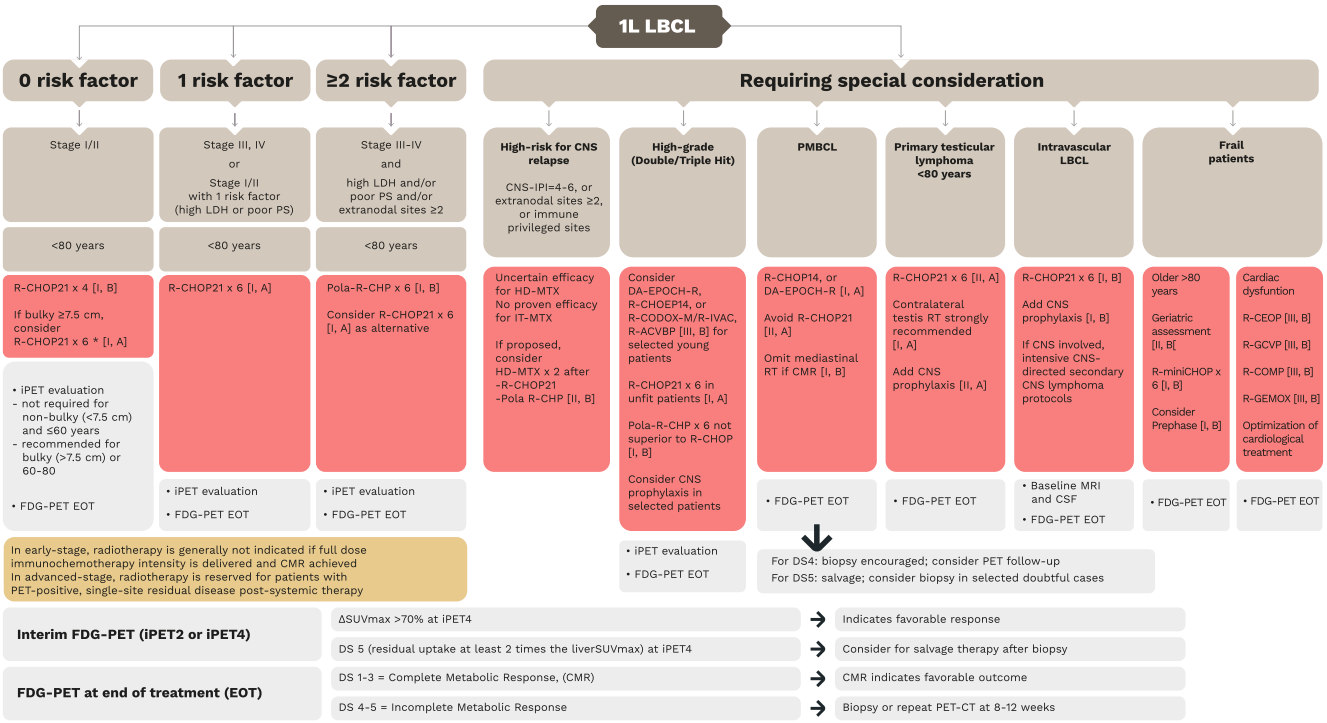

FIGURE 1 **Algorithm 1L large B-cell lymphoma (LBCL) treatment.** CNS, central nervous system; CT, computed tomography; DA-EPOCH-R, dose-adjusted etoposide, doxorubicin, vincristine, cyclophosphamide, prednisone, rituximab; DS, Deauville Score; EOT, end of treatment; FDG-PET, fluorodeoxyglucose-positron emission tomography; HD-MTX, high-dose intravenous methotrexate; iPET, interim PET at 2, 3 or 4 cycles depending on local practice; IT-MTX, intrathecal methotrexate; LDH, lactate dehydrogenase; MRI, magnetic resonance imaging; PET, positron emission tomography; PMBCL, primary mediastinal B-cell lymphoma; Pola-R-CHP, polatuzumab vedotin rituximab, cyclophosphamide, doxorubicin, and prednisone; PS, performance status; R-ACVBP, rituximab, doxorubicin, cyclophosphamide, vindesine, bleomycine, and prednisone; R-CEOP, rituximab, cyclophosphamide, etoposide substituted for doxorubicin, vincristine, prednisone; R-CHOEP-14, R-CHOP + etoposide every 14 days; R-CHOP, rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone; R-CHOP14, R-CHOP given every 14 days; R-CHOP21, R-CHOP given every 21 days; R-CODOX-M/R-IVAC, rituximab doxorubicin vincristine cyclophosphamide, IV methotrexate/etoposide, ifosfamide (with MESNA), cytarabine Intrathecal cytarabine intrathecal methotrexate; R-COMP, RCHOP with non-PEGylated liposomal doxorubicin; R-GCVP, rituximab, cyclophosphamide, vincristine, gemcitabine, and prednisolone; R-GEMOX, rituximab, gemcitabine-oxaliplatin; RT, radiotherapy; SUV standardized uptake value.

Summary of recommendations 1L LBCL

General considerations.

- Bulky disease: defined as a maximal tumor diameter of ≥7.5 cm[52]; a more accurate definition is based on metabolic volume when TMTV > 220 mL or included in the IMPI as a continuous variable.[39,40]
- In early stage (Stage I/II), low-risk or high-risk disease is defined based on age (≤60 years old, >60 years old) and the presence of adverse clinical factors (high LDH and poor PS).
- RT (ISRT 30 Gy in 15 fractions): In the early stage, RT is generally not indicated for chemosensitive disease if full dose intensity is delivered because it offers limited survival benefit[58,59] **[III, C]**. In the advanced stage, consolidation in the case of an FDG-positive residual lesion in a single site after systemic therapy may be proposed **[II, B]**.
- Assessment by FDG-PET at EOT is mandatory.

1L LBCL, Stage I/II, 0 risk factor, <80 years.

- Offer R-CHOP21 × 4 cycles (IB).
  - **Bulky:** consider R-CHOP21 × 6 cycles (IA).
  - **Non-bulky and <60 years:**

Response assessment:

* Interim PET evaluation:
  - Not required for non-bulky (<7.5 cm) and ≤60.
  - Recommended for bulky (≥7.5 cm) or 60–80.
* FDG-PET at EOT for all patients.

1L LBCL, 1 risk factor (Stage III/IV or Stage I/II with poor PS or high LDH), <80 years.

- offer R-CHOP21 × 6 cycles (**1A**).

Response assessment:

* Interim PET evaluation: recommended.
* FDG-PET at EOT for all patients.

1L LBCL, Stage III/IV, ≥2 risk factors, <80 years.

- offer Pola-R-CHP × 6 cycles **[I, B];** consider R-CHOP21 as an alternative **[I, A]**.

Response assessment:

* Interim PET evaluation: recommended.
* FDG-PET at EOT for all patients.

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TABLE 6 Detailed acronyms of chemotherapies.

Acronyms	Drugs
R-CHOP	Rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone
R-CHOP21	R-CHOP given every 21 days
R-CHOP14	R-CHOP given every 14 days
Pola-R-CHP	Polatuzumab vedotin rituximab, cyclophosphamide, doxorubicin, and prednisone
DA-EPOCH-R	Etoposide, doxorubicin, vincristine, cyclophosphamide, prednisone, and rituximab Dose-adjusted based on the neutrophil nadir, which is checked twice weekly
R-CHOEP-14	R-CHOP + etoposide every 14 days
R-CODOX-M/R-IVAC	Rituximab doxorubicin vincristine cyclophosphamide, IV methotrexate/etoposide, ifosfamide (with MESNA), cytarabine Intrathecal cytarabine intrathecal methotrexate
R-ACVBP plus sequential consolidation	Rituximab, doxorubicin, cyclophosphamide, vindesine, bleomycine, and prednisone, followed with consolidation with 2 cycles of HD. MTX (3 g/m^2), 4 cycles of rituximab, etoposide, and ifosfamide, followed by 2 cycles of cytarabine subcutaneously for 4 days
R-CEOP	Rituximab, cyclophosphamide, and etoposide substituted for doxorubicin, vincristine, and prednisone
R-GCVP	Rituximab, cyclophosphamide, vincristine, gemcitabine, and prednisolone
R-COMP	RCHOP with non-PEGylated liposomal doxorubicin
R-GEMOX	Rituximab, gemcitabine-oxaliplatin
R-DHAP /X/ or C	Rituximab, dexamethasone, cytarabine, and cisplatin or oxaliplatin or carboplatin
R-ICE	Rituximab, ifosfamide, carboplatin, and etoposide
R-GDP	Rituximab, gemcitabine, dexamethasone, and cisplatin
R-ESHAP	Rituximab, etoposide, cisplatin, and methylprednisolone
BEAM	Carmustine/BCNU, etoposide, cytarabine, and melphalan
BEAC	Carmustine/BCNU, etoposide, cytarabine, and cyclophosphamide
BeEAM	Bendamustine + etoposide + cytarabine + melphalan
FEAM	Fotomustine + etoposide + cytarabine + melphalan
Pola-BR	Polatuzumab vedotin, bendamustin, and rituximab

LBCLs requiring special consideration

Patients at high risk for CNS relapse

Relapse of LBCL in the CNS is rare but often fatal.[72] Prophylaxis methods include IT injections and high-dose intravenous methotrexate (HD-MTX), although their efficacy is debated,[73,74] and they yield significant toxicity, especially in elderly patients.

Risk factors for CNS relapse

- CNS-IPI ≥4–6 (10%–12% risk).[31]
- Two or more than two initial extra-nodal sites.[75]
- Involvement of specific extranodal sites, named immune privileged sites: testis, breast, epidural space, orbit, kidney, and adrenal gland.[76]

High-grade histology—particularly in lymphomas expressing the dark-zone gene expression signature (DZsig), originally derived from HGBCL-DH-BCL2—as well as MYC rearrangements and genetic subtypes involving MYD88/CD79 mutations, have been suggested to be associated with an increased risk of CNS relapse.[77] However, the strength of these associations remains uncertain, and further validation is needed before these features can be considered reliable risk factors in clinical practice.

Initial screening.

- Routine screening in high-risk patients should include MRI and CSF analysis via cytology and flow cytometry—which demonstrated a greater sensitivity than cytology- or cell-free tumor DNA (ctDNA), the latter being a promising method not yet widely available nor standardized.[12]
- Perform these examinations as early as feasible after diagnosis **[III, A]**, ideally before Cycle 1.

CNS prophylaxis strategies

IT methotrexate has limitations in penetration and efficacy and is not recommended. HD-MTX with at least 3 g/m^2 × 2 cycles may be considered for patients with high CNS relapse risk who have obtained a CMR after R-CHOP or Pola-R-CHP **[III, B]**. Intensive regimens such as R-CHOEP, rituximab doxorubicin vincristine cyclophosphamide, IV methotrexate/etoposide, ifosfamide (with MESNA), cytarabine-intrathecal cytarabine intrathecal methotrexate (R-CODOX-M/R-IVAC), or rituximab, doxorubicin, cyclophosphamide, vindesine, bleomycine, and prednisone (R-ACVBP), followed by sequential consolidation—which incorporate HD-MTX[78]—may be considered in selected young and fit patients under 60 years of age **[III, B]**. However, the supporting evidence remains limited and is primarily based on retrospective analyses.

Recommendations

- CNS-IPI should be assessed for all patients with LBCL **[III, A].**
- Perform MRI and CSF analysis for high-risk patients as early as feasible after diagnosis **[III, A].**

High-grade B-cell lymphoma with translocations of *MYC* and *BCL2* and/or *BCL6* (double- and triple-hit lymphomas, DH/TH)

Patients with *MYC*, *BCL2*, and/or *BCL6* translocations and disseminated disease experience poorer outcomes with R-CHOP21. R-CHOP21 should be considered only for unfit patients for whom intensive regimens are not feasible. CNS prophylaxis may be warranted only in selected patients. For eligible patients, intensive regimens like dose-adjusted etoposide, doxorubicin, vincristine, cyclophosphamide, prednisone, rituximab (DA-EPOCH-R), R-CHOEP-14, R-CODOX-M/R-IVAC, and R-ACVBP are recommended **[III, B]**.[79]

For these groups, Pola-R-CHP does not appear to be superior to R-CHOP.[12]

HD-CT with ASCT as consolidation after first-line treatment is not recommended **[IV, B]**.[80]

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Recommendations

- **High grade (double/triple hit):** Consider R-CHOEP14 or DA-EPOCH-R, R-CODOX-M/R-IVAC, and R-ACVBP for selected patients **[III, B]**. Pola-RCHP seems not superior to R-CHOP in D/TH LBCL. Offer R-CHOP21 only in unfit patients.

Primary mediastinal B-cell lymphoma

Primary mediastinal B-cell lymphoma (PMBCL) primarily affects young females and is characterized by a bulky anterior mediastinal mass, rapid progression, and a distinct gene expression profile.[81]

- Chemoimmunotherapy regimens

  Long-term outcomes with regimens such as R-CHOP14 or more intensified regimens including R-M/VACOP-B, R-ACVBP, and DA-EPOCH-R for patients <80 are generally favorable and show comparable complete response (CR) rates and long-term outcomes.[82–85] R-CHOP21 is associated with inferior results and should be avoided in PMBCL **[II, A]**.[82,84–87]
- RT

  In the IELSG37 Phase 3 trial,[85] 545 patients were treated with various regimens, including R-CHOP21 and DA-EPOCH-R. Among patients achieving CMR (DS 1–3, 50% of patients), no significant difference was observed between the observation and RT arms in terms of the 30-month PFS (96.2% vs. 98.5%) or the 5-year OS (99%).

  Mediastinal RT can be omitted for patients achieving CMR, improving long-term safety **[II, B]**.
- Management of DS4/DS5 cases

  For patients with DS4 at the end of chemoimmunotherapy, biopsy is encouraged. A close follow-up with FDG-PET may be proposed, namely, after DA-EPOCH-R. These patients may also benefit from RT, with a 5-year PFS of 95.8% in the IELSG37 trial **[III, C]**.[85] RT may be considered in the context of long-term complications and potential excellent results of novel therapies in Line 2, such as CAR-T therapy and checkpoint inhibitors, appearing more appropriate and effective for these patients with supporting data already reported **[III, C]**.[88]

  DS5 cases have a worse prognosis, underscoring the need for salvage therapies **[III, B]**.

Recommendation for PMBCL ≤80

- Treat with R-CHOP14 or more intensified regimens including R-M/VACOP-B, R-ACVBP, and DA-EPOCH-R for patients <80 **[I, A]**.
- Omit mediastinal RT in patients achieving CMR after chemoimmunotherapy **[I, A]**.
- Avoid R-CHOP21 in favor of regimens with better outcomes **[II, A]**.
- For DS4 cases, biopsy should be encouraged; consider follow-up with FDG-PET, namely, after DA-EPOCH-R. Consider RT evaluating long-term complications **[II, B]**.
- DS5 cases require salvage treatment **[I, B].** Consider biopsy in selected doubtful cases.

Primary testicular LBCL

Primary testicular LBCL (PTL) accounts for less than 2% of non-Hodgkin lymphomas and predominantly involves the ABC-like subtype. PTL often presents as limited-stage disease, with unilateral testis involvement (Stage I) ± para-aortic lymph nodes (Stage II).

Without contralateral testis RT, PTL carries a high risk of relapse in the contralateral testis, extranodal sites, and the central nervous system (CNS).[89,90]

The IELSG10 Study, including patients <80, showed significant outcome improvement with R-CHOP21 (6–8 courses) combined with CNS prophylaxis (four doses of intrathecal [IT] methotrexate) and contralateral testis RT **[III, B]**.[91] The study achieved 5-year PFS and OS of 74% and 85%, with no contralateral testis relapses and a 6% CNS relapse rate.

In the more recent IELSG30 study,[92] intensified CNS prophylaxis with IT liposomal cytarabine and HD-MTX was implemented; regional lymph node RT was omitted, achieving a 5-year PFS of 88% with no CNS or contralateral testis relapses. Emerging evidence suggests that systemic IV HD-MTX may be more effective than IT chemotherapy for CNS prophylaxis.

Recommendations

- Offer R-CHOP (6 cycles) **[II, A]**.
- Consider as strongly recommended contralateral testis RT following chemoimmunotherapy **[I, A]**.
- Offer CNS prophylaxis **[II, A]**.
- Treat advanced-stage PTL similarly to nodal LBCL, but include prophylactic testis RT and CNS prophylaxis **[III, B]**.

Intravascular LBCL

Intravascular LBCL is a rare LBCL, characterized by neoplastic B cells within the lumen of blood vessels, more commonly affecting older patients and often involving extranodal sites such as the CNS (approximately 30% of patients), lungs, and skin.[93] The prognosis is generally poor. Rituximab may be associated with severe infusion-related reactions.[94] This may be mitigated during initial induction by delaying rituximab until 2–3 days after the first dose of chemotherapy. Seeking to mitigate the high risk of CNS involvement, an early-phase trial incorporating IT and HD-MTX into an R-CHOP backbone reported a 2-year PFS of 76%.[95]

Recommendations

- Perform contrast-enhanced MRI of the brain (include the spine if clinically indicated) and baseline CSF assessment (to include cytology and immunophenotyping) as baseline screening for CNS disease (1B).
- Offer six cycles of R-CHOP (1B).
  - Consider delaying the first rituximab infusion by ≥48 h after chemotherapy administration to mitigate against the risk of a severe infusion reaction in Cycle 1 (2B).
- Offer CNS prophylaxis (see chapter) for those with no evidence of CNS disease at baseline (1B).
- Offer intensive CNS-directed lymphoma protocols for those with evidence of CNS disease (1B).

Older patients

Older patients are usually defined as individuals aged ≥80 years old or unfit patients <80 years old. A thorough pre-therapeutic assessment is essential for elderly patients, considering their overall health, comorbidities, preferences, and lymphoma characteristics (Table 7). Diagnosis and staging should align with practices for younger patients. Simplified geriatric assessment or other frailty indices are

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TABLE 7 Geriatric assessment.

Category	Assessment tool	Focus
Functional status	ECOG performance scale ADL, IADL	Physical independence
Comorbidities	CIRS-G	Chronic conditions and severity
Cognitive function	MMSE or MoCA	Memory and executive functioning
Nutritional status	BMI, albumin	Weight loss and malnutrition risk
Social support	Geriatric Depression Scale	Mental health and caregiver support
Polypharmacy	Medication review	Risk of drug interactions

Abbreviations: ADL, activities of daily living; BMI, body mass index; IADL, instrumental activites of daily living; MMSE, mini mental state of examination; MoCA, montreal cognitive assessment.

recommended to evaluate cognitive function, nutritional status, and frailty, distinguishing fit, unfit, and frail patients.[96] This helps identify candidates for curative treatments versus those needing attenuated approaches or palliative care. Goals of care discussions with patients and families should include palliative options when appropriate. Optimizing all aspects of supportive care is important to reduce morbidity, particularly for elderly or frail patients. A palliative approach that focuses on quality of life and symptom control may be more suitable for very frail patients.

Two large Phase II trials validate six cycles of R-miniCHOP (doxorubicin 25 mg/m^2, cyclophosphamide 400 mg/m^2, and vincristine 1 mg) for patients ≥80, achieving curative potential **[I, B]**.[97] Prephase corticosteroids with or without vincristine improve PS and mitigate subsequent immunochemotherapy's adverse effects **[I, A]**.[98–100]

Patients with cardiac dysfunction **left ventricular ejection fraction < 50% or signs of heart failure** necessitate doxorubicin-free regimen, including rituximab, cyclophosphamide, and etoposide substituted for doxorubicin, vincristine, and prednisone (R-CEOP) **[III, B]**, rituximab, cyclophosphamide, vincristine, gemcitabine, and prednisolone (R-GCVP), or non-PEGylated liposomal doxorubicin (R-COMP), rituximab, gemcitabine-oxaliplatin (R-GEMOX),[101–105] appropriate management of the cardiac function including optimization of cardiological treatment (ACE inhibitors, beta-blockers), and echo heart monitoring every two cycles (+troponin and NTproBNP as per cardio-oncology advice).

In case of previous treatment with anthracyclines (e.g., breast cancer, FEC), calculate the anthracycline dose limit (450 mg/m^2) and make decisions accordingly.

Osteoporosis risk is significant in older patients receiving R-CHOP-like therapy,[45,106] warranting evaluation using the FRAX score and considering vitamin D and bisphosphonates. Participation in clinical trials with chemo-free regimens is encouraged.

Recommendations for patients aged ≥80 or unfit ≤80

- Use geriatric assessment tools to optimize therapy **[II, B]**.
- Assess and manage osteoporosis risk **[I, B]**.
- Consider a prephase corticosteroids to improve performance **[I, B]**. A single intravenous 1 mg dose of vincristine can be added to corticosteroid prephase **[II, B]**.
- Offer R-mini-CHOP × 6 cycles for fit patients ≥80 **[I, B]**.
- Offer non-anthracycline-based regimens for patients with cardiac dysfunction or signs of heart failure **[I, B]**, with appropriate management of cardiac function.
- Participation in clinical trials with chemo-free regimens is encouraged.
- Patient preferences should be considered in treatment decisions.

REFRACTORY/RELAPSED LBCL

Although the majority of patients can be cured with 1 L immunochemotherapy, approximately 30%–40% will have primary refractory LBCL or relapse of LBLCL after initial remission. Primary refractory disease is defined as SD or PD during, or by the end of 1 L treatment.[107] Relapse after an initial response may be early (within 12 months of completing immunochemotherapy) or late (>12 months). This impacts the prognosis and treatment strategy.

Emotional distress assessment should at a minimum be performed at every treatment milestone, including relapse. It is also important to recheck patient preferences, as they may have changed since initial therapy.

Diagnostic approach to refractory/relapsed disease

Before starting 2L, biopsy should be performed **[III, B]**.[108,109] This is to confirm that the physical findings/imaging results are indeed due to relapsed LBCL rather than other malignancies, an alternative lymphoma histology requiring a distinct approach, or another diagnosis, such as sarcoid-like reaction.[19] Additionally, it is important to determine the expression of targetable antigens such as CD19 and CD20.

Patient fitness and time from relapse should be assessed to select the appropriate treatment approach.

2L therapy

At the time of relapse, refractory disease or relapse is defined by the persistence or reappearance of lymphoma. During treatment, the presence of only a partial metabolic response assessed by PET-CT is considered by most clinicians as treatment failure, and biopsy should be performed to confirm the refractoriness.

Primary refractory or relapse within 12 months of completing the first-line treatment

In transplant-eligible patients with primary refractory disease or relapse within 12 months of completing first-line treatment, two of three randomized-controlled trials[110–114] have demonstrated a clinically and statistically significant benefit when treated with CAR T-cells, axicabtagene ciloleucel (axi-cel), or lisocabtagene maraleucel (liso-cel). Using axi-cel, after a median follow-up of 24.9 months, the estimated event-free survival (EFS) at 2 years was 41% versus 16%, respectively (hazard ratio [HR], 0.40; 95% confidence interval [CI], 0.31–0.51, P < 0.001).[110] Furthermore, in a recent follow-up update (median 47.2 months), better OS was also observed in the axi-cel arm, with 4-year estimates of 55% versus 46%, respectively (HR, 0.73; 95% CI, 0.54–0.98; P = 0.03) **[I, A]**.[113] Using liso-cel, after a median follow-up of 17.5 months, the median EFS had not been reached in the liso-cel arm and was 2.4 months in the standard treatment arm (HR, 0.36; 95% CI, 0.2–0.52, P < 0.001). A trend toward better OS was also observed in the liso-cel arm (73% vs. 61% at 18 months), although the difference was not statistically significant (HR, 0.72; 95% CI, 0.44–1.18; P = 0.0987) **[I, A]**.[112,115] Clinical trials have also been carried out with liso-cel (PILOT)[116] and axi-cel (ALYCANTE)[115] as the second-line treatment in patients who are not candidates for

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transplant due to their age, comorbidities, or compromised organ function.

Relapse >12 months of completing the first-line treatment and fit for intensive therapies

Patients fit for intensive therapies are typically under the age of 70 without significant comorbidities/organ dysfunction. Fit patients over the age of 70 may derive benefit, but they may have a slower functional recovery.

The standard approach consists of reinduction immunochemotherapy and stem cell collection, followed in patients with CMR by HD-CT and ASCT.[117] This results in long-term PFS rates of around 40% in late relapsing patients, but less than 15%–20% in primary refractory or early relapsing patients. Commonly used regimens include R-DHAP, R-ICE, R-GDP, and R-ESHAP. Randomized studies and comparison of outcomes of Phase II data suggest that these regimens have similar efficacy,[118] but R-GDP has the advantage that it can be delivered in an outpatient setting.[118] Patients in CMR after two cycles can proceed to HD-CT with carmustine/BCNU, etoposide, cytarabine, and melphalan (BEAM) or variants such as carmustine/BCNU, etoposide, cytarabine, and cyclophosphamide (BEAC)[119] or bendamustine + etoposide + cytarabine + melphalan (BeEAM)[120] or fotomustine + etoposide + cytarabine + melphalan (FEAM).[121]

Patients not eligible for CAR T-cell therapy (relapse <12 months) or patients not eligible for HD-CT/ASCT (relapse >12 months)

Significant challenges remain in the use of CAR-T therapy. First, given the need for rapid treatment in patients with aggressive chemo-refractory disease, delays in product manufacturing may affect patient eligibility, so "bridging" therapy with steroids, RT, or immunochemotherapy is needed in most cases to control the disease until the product can be infused.[122–124] Patients whose general condition has deteriorated (PS ≥ 2) or who have a large tumor volume or and rapidly increasing LDH levels have worse outcome with CAR-T therapy.[122–126]

Immunochemotherapy such as R-GEMOX[127] can be considered. Tafasitamab-lenalidomide[128–130] is approved in this setting. However, patients with refractory disease were primarily excluded from the pivotal trial. Moreover, the outcome of early relapsed (<6 months) disease in real-life series was worse than reported in the original study.[131] Polatuzumab vedotin (Pola-BR) is an effective regimen.[132] Bendamustine might impair T-cell fitness and could hamper the effectivity of epcoritamab, glofitamab, and odronextamab, which could be used in the third line. For patients with limited disease, ISRT is an option. For some patients, best supportive care should be discussed.

Recently approved, the combination of the bispecific antibody glofitamab with gemox (Glofi-gemox) was shown to be superior to R-gem-ox, with a superior OS (median 25.5 months [18.3-NE] vs. 12.9 months [7.9–18.5]; HR 0.62 [0.43–0.88]) and PFS, with a median follow-up of 20.7 months (19.9–23.3).[133] The recommendation is to consider bispecific glofitamab + gemox for patients ineligible for ASCT. Epcoritamab + gemox is under investigation[134] and should also be considered for patients ineligible for HDT/ASCT when approved, as well as Pola-R-gemox.

Summary of recommendations

- Before starting 2L treatment, a biopsy is strongly recommended **[III, B]**.
- Patients with refractory disease or relapse within 12 months of completion of first-line treatment: offer axicabtagene ciloleucel **[I, A]** or lisocabtagene maraleucel **[I, A]**.
- CAR T-cell therapy may not be appropriate in patients with PS ≥ 2 or who have a large tumor volume and/or rapidly increasing LDH levels.[122,125]
- Patients who relapse >1 year of completion of 1L therapy and who are fit for intensive therapy: offer immunochemotherapy (R-DHAX [P or C], R-ICE, R-GDP, R-ESHAP) and in case of chemosensitivity (CMR or very good PR = HD-CT [BEAM, BEAC, BeEAM, and FEAM], followed by ASCT **[I, B]**.
- Patients not eligible for CAR T-cell therapy (relapse < 12 months) or patients not eligible for HD-CT/ASCT (relapse > 12 months): offer R-gem-ox or pola-BR **[III, B]**.
- Consider glofitamab + GemOx **[I, A]** or epcoritamab + GemOx **[III, C]** when approved and reimbursed for patients ineligible for ASCT **[III, C]**.
- Consider tafasitamab-lenalidomide in non-refractory patients **[III, C]**.
- In limited-stage disease: consider palliative radiation (IS-RT).
- In unfit patients: consider best supportive care.

3L and beyond

The different routes to third-line therapy and beyond require a tailored approach to patient management, based on prior treatment and patient preference. Biopsy is highly recommended prior to 3L particularly, given high rates of target antigen loss that will increasingly guide treatment decisions.

Anti-CD19 CAR-T therapies (axi-cel, tisa-cel, and liso-cel) are approved for R/R LBCL after two or more prior lines of therapy based on single-arm pivotal trials, given the high ORRs of 52%–83% with CRRs between 40% and 53%.[135–137] Responses have proven to be durable in approximately 40% of patients.[138–140] These findings have been confirmed through the reporting of "real-word" experience from multiple consortia.[130,141–145] Product choice may be guided by differential availability across territories and toxicity profiles. Consideration should be given to optimal bridging strategies.[124] Selection of the patients for CAR-T is recommended based on PS and evaluation of the aggressivity of the tumor (e.g., large tumor volume or rapidly increasing LDH levels).

CD20-CD3 bispecific antibodies are proving to be highly efficacious in the treatment of R/R LBCL. Three molecules, glofitamab, epcoritamab, and odronextamab, are approved in R/R LBCL patients after two prior lines of therapy.[146–148] CR rates of around 40% were described in these intensively pretreated and mostly refractory patients, with no significant differences with respect to prior exposure to CAR-T cell therapy. The remissions seem long-lasting, since most patients who achieve CR maintain it for at least 1 year.

Additional recently introduced potential therapeutic agents include loncastuximab tesirine, an antibody drug conjugate targeting CD19 with a pyrrolobenzodiazepine payload,[149] and tafasitamab, a CD19-directed humanized antibody, in combination with lenalidomide.[128]

In all patients, the choice of therapy should be based on shared decision-making, taking into account the patient's wishes in terms of life goals, mode/frequency of delivery, duration of therapy, and toxicity. Palliative chemotherapy/steroids can be considered, including combinations that may include etoposide, cyclophosphamide, lomustine, or procarbazine.[150–153] Also, palliative RT and best supportive care should be discussed.

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Summary of recommendations

- Available options for patients with R/R LBCL after ≥2 prior lines of therapy are multiple and the choice of therapy should be tailored based on patient wishes in terms of life goals, comorbidities, ECOG PS, previously used regimens, duration of therapy and toxicities, and mode/frequency of delivery.
- These options are as follows:
  * if not treated with anti-CD19 CAR T-cells in the second line: offer CD19 CAR T-cell therapy (axi-cel **[III, A]**, tisa-cel **[III, A]**, or liso-cel **[III, A]**).
  * if CAR-exposed: offer CD20xCD3 bispecific antibody (epcoritamab **[III, C]**, gloflitamab **[III, C]**, odronextamab **[III, C]**), loncastuximab tesirine **[III, C]**, tafasitamab-lenalidomide **[III, C]**, or an immunochemotherapy regimen.
- Consider the patient for a clinical trial.
- In limited-stage disease: consider palliative radiation (IS-RT).
- In unfit patients: consider best supportive care (Figure 2).

FOLLOW-UP, LONG-TERM IMPLICATIONS, AND SURVIVORSHIP

Patients with DLBCL who are event-free at 2 years after diagnosis have an identical OS to that of the general population, emphasizing the need to only specifically monitor the disease in this early period.[154]

Patients should be aware of possible symptoms or relapse/progression and urgent contact details in these occurrences. An EoT consultation should be offered to every patient. This should include an EoT Holistic Needs Assessment (HNA) and associated written care plan and should also include the discussion and provision of a comprehensive treatment summary. On successful completion of treatment, both the patient and their general practitioner (GP) should be made aware of follow-up plans and potential future disease or treatment-related issues.

Careful history and physical examination for at least 2 years or longer with attention to development of secondary tumors or other long-term side effects of treatment is recommended based on the cumulative dose and anthracyclines and radiation exposure **[III, C]**.

Blood count should be carried out at least for 2 years, and then only as needed for evaluation of suspicious symptoms or clinical findings in those patients suitable for further therapy **[III, C]**.

Routine surveillance PET scans are not recommended due to the lack of evidence for outcome improvement.[155–157] Inconclusive positive PET results should prompt biopsy or repeat PET-CT where feasible.[47,158] Patients with LBCL who are event-free at 2 years have an OS comparable to that of the general population, highlighting the importance of focused disease monitoring during this early period.[154] For high-risk patients eligible to chimeric antigen receptor T (CART), a PET-CT may be repeated within 6–12 months after the EoT. After 5 years of relapse-free follow-up, patients can be considered cured.

Long-term side effects of chemotherapy, such as anthracycline-induced cardiotoxicity, require monitoring.[159,160]

Echocardiography and cardiology referrals are recommended for all patients (anthracyclines, CAR-T, autograft, and bispecifics)

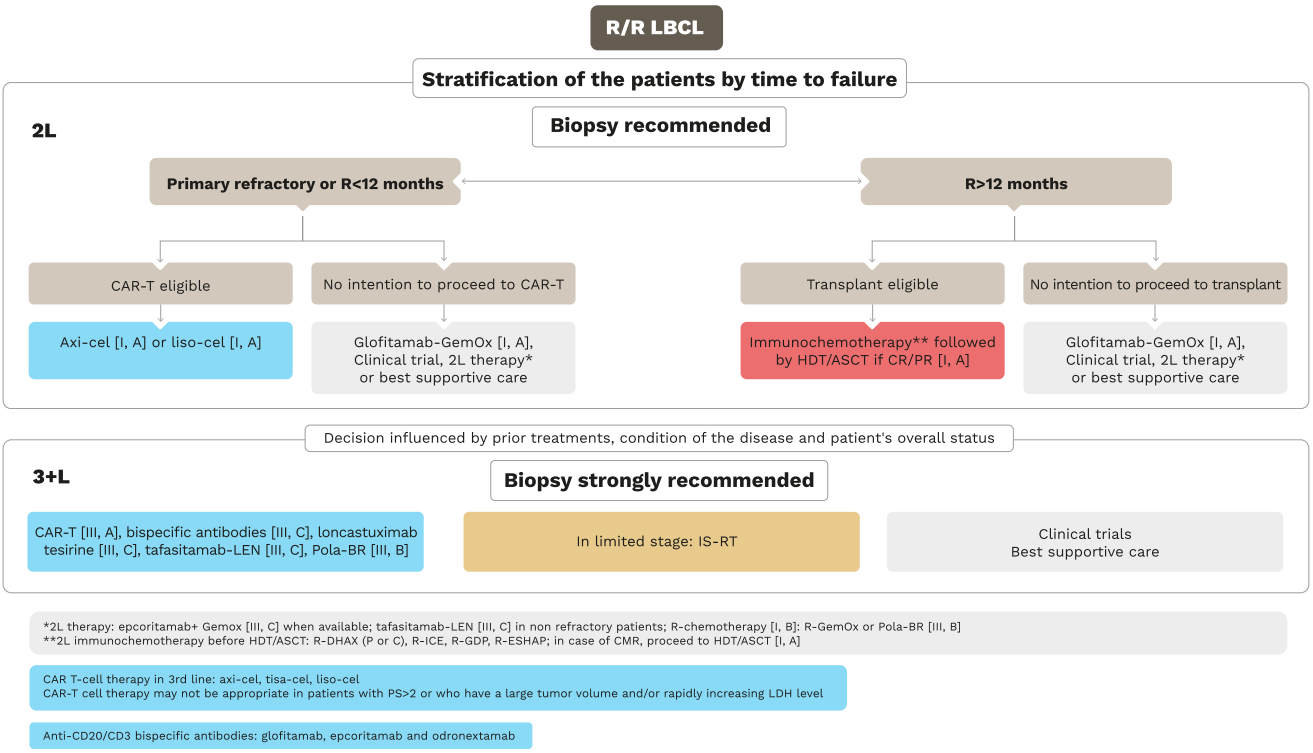

FIGURE 2 **Algorithm.** Treatment for large B-cell lymphoma (LBLCL) in 2L and 3+L. ASCT, autologous stem cell transplant; CMR, complete metabolic response; CR, complete response; HDT, high - dose therapy; IS-RT, involved site radiotherapy; LDH, lactate dehydrogenase; Pola-BR, polatuzumab vedotin, bendamustin, rituximab; PR, partial response; PS, performance status; R-DHAX (P or C), rituximab, dexamethasone, cytarabine, and oxaliplatin (cisplatin or carboplatin); R-ESHAP, rituximab, etoposide, cisplatin, and methylprednisolone; R-GDP, rituximab, gemcitabine, dexamethasone, and cisplatin; R-GemOx, rituximab, gemcitabine-oxaliplatin; R-ICE, rituximab, ifosfamide, carboplatin, and etoposide.

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with a cardiology consultation 1 year after the EoT, including ECG, lipid profile, HbA1C, and echocardiography. Particularly, for patients with fewer than three cardiovascular risk factors, echocardiography at the end of anthracycline treatment and for patients with more than three cardiovascular risk factors, echocardiography at the end of anthracycline treatment and at 3 months is recommended. During long-term follow-up, for patients who received ≥300 mg/m^2 of doxorubicin, or doxorubicin combined with mediastinal RT, or those with underlying heart disease or who experienced cardiotoxicity, annual screening for cardiovascular risk factors including ECG + echocardiogram (ECHO) at 3 and 5 years, and then every 5 years is recommended.

Include monitoring for long-term immune dysfunction, as patients remain at increased risk of infections posttreatment. Preventative strategies such as vaccinations and antimicrobial prophylaxis should be incorporated when indicated.

Polyneuropathy should be addressed early with treatments like gabapentin and duloxetine.

Smoking cessation counseling is essential for all patients.

Many patients suffer from fatigue or anxiety.[161] Therefore, routine assessments for these should be integrated into follow-up care, with referrals to psychological and rehabilitation support services as needed to address mental distress and fear of recurrence. Psychological distress may still occur and may be life-impacting for the patient.

Distress screening should at a minimum be performed at every treatment milestone, including when the patient is in survivorship follow-up, to ensure timely referral to support as needed.[162]

Evidence-based lifestyle interventions, such as physical activity and a healthy diet, should be recommended to manage fatigue, improve cardiovascular health, and support long-term recovery.[162]

Secondary tumors are rare in DLBCL due to the median age of presentation but warrant attention in younger patients exposed to mediastinal RT, with breast cancer screening recommended starting 8 years after the end of RT, consisting of a single oblique incidence mammogram without tomosynthesis, complemented by ultrasound and breast MRI [5D]. From the age of 50, mammograms should be performed every 2 years. TSH testing has to be checked at 1, 2, and 5 years if neck irradiation was performed.[163]

Summary of recommendations

- Offer an EoT consultation to every patient.
- Avoid routine surveillance CT scans or PET-CT scans in asymptomatic patients (**3B**). For high-risk patients eligible for CART, PET-CT may be performed within 6–12 months from the EoT.
- Careful history and physical examination at least for 2 years, with attention to development of secondary tumors or other long-term side effects of chemotherapy, is recommended **[III, C]**.
- Blood count should be carried out at least for 2 years, and then only as needed, for evaluation of suspicious symptoms or clinical findings **[III, C]**.
- Cardiology referrals and echocardiogram surveillance are recommended within the 1-year posttreatment, then at 3 years and at 5 years, and then every 5 years.
- Polyneuropathy should be addressed early.
- Fatigue can be persistent. Screening is warranted to ensure that there is no underlying cause. Patients should also be given credible, understandable information on cancer-related fatigue and coping mechanisms.
- Psychological concerns like anxiety and fear of recurrence are frequent. Refer patients for psychological support as needed.
- Smoking cessation, breast cancer screening.

AUTHOR CONTRIBUTIONS

Catherine Thieblemont: Conceptualization; writing—original draft; validation; supervision. **Maria Gomes Da Silva**: Conceptualization; writing—original draft. **Sirpa Leppä**: Conceptualization; writing—original draft. **Georg Lenz**: Conceptualization; writing—original draft. **Anne-Ségolène Cottereau**: Conceptualization; writing—original draft. **Christopher Fox**: Conceptualization; writing—original draft. **Armando Lopez-Guillermo**: Conceptualization; writing—original draft. **Timothy Illidge**: Conceptualization; writing—original draft. **Wojciech Jurczak**: Conceptualization; writing—original draft. **Hans Eich**: Conceptualization; writing—original draft. **Igor Aurer**: Conceptualization; writing—original draft. **Marek Trneny**: Conceptualization; writing—original draft. **Andreas Rosenwald**: Conceptualization; writing—original draft. **Andrew Davies**: Conceptualization; writing—original draft. **Ben (Gerben) Zwezerijnen**: Conceptualization; writing—original draft. **Natacha Bolanos**: Conceptualization; writing—original draft. **Maja Marković**: Conceptualization; writing—original draft. **Jean-Philippe Jais**: Methodology. **Florence Broussais**: Methodology; project administration; writing—review and editing. **Martin Dreyling**: Supervision. **Umberto Vitolo**: Conceptualization; writing—original draft. **Hervé Tilly**: Conceptualization; writing—original draft. **Marie José Kersten**: Conceptualization; writing—original draft; supervision.

CONFLICT OF INTEREST STATEMENT

Catherine Thieblemont discloses a direct financial relationship (honoraria) with Janssen, Roche, AbbVie, Novartis, BMS, Incyte, and BeiGene. Maria Gomes Da Silva discloses financial and/or professional relationships with Janssen Cilag (Direct and Indirect), Roche (Direct), Lilly (Direct and Indirect), AstraZeneca (Indirect), Gilead (Direct), BeiGene (Indirect), AbbVie (Direct and Indirect), and Takeda (Direct), and professional relationships with Sociedade Portuguesa de Hematologia. Sirpa Leppä discloses a direct financial relationship with AbbVie, Genmab, Incyte, Roche, and Sobi, and an indirect financial relationship with Bayer, BMS, Genmab, Hutchmed, Novartis, and Roche. Georg Lenz discloses a direct financial relationship with Roche/Genentech, Gilead, BMS, Novartis, AstraZeneca, AbbVie, Incyte, PentixaPharm, Sobi, Immogene/FLINDR, Hexal/Sandoz, Lilly, BeiGene, and MSD, and an indirect financial relationship with Roche/Genentech, Gilead, Bayer, Novartis, AstraZeneca, AbbVie, MSD, and Pierre Fabre. Armando Lopez-Guillermo discloses a direct financial relationship with AbbVie, Atarabio, BMS, GenMab, Gilead/Kite, Incyte, Janssen, Lilly, Morphosys, Ono, Roche, SERB, SOBI, and Takeda, and an indirect financial relationship with BeiGene and AbbVie. Timothy Illidge discloses a direct financial relationship with Takeda. WJ discloses a direct and indirect financial relationship with AbbVie, AstraZeneca, BeiGene, Janssen Cilag, Lilly, Regeneron, and Roche. Hans Eich discloses a direct financial relationship (honoraria) with Kyowa Kirin and Takeda and serve as a Steering Committee Member of the International Lymphoma Radiation Oncology Group (ILROG). Igor Aurer discloses a direct financial relationship with Roche, Novartis/Sandoz, AbbVie, AstraZeneca, Genesis/Incyte, Sobi, and Takeda. Marek Trneny discloses a direct financial relationship with Roche, Gilead, AstraZeneca, Sobi, Takeda, Incyte, Lilly, BMS, Novartis, AbbVie, Janssen, Swixx, Carbou Sciences, and Autolus. Andrew Davies discloses a direct financial relationship with Roche (conference attendance, honorarium for talks, advisory boards), AstraZeneca, Kite/Gilead, Sobi, AbbVie, Genmab, Celgene, MSD, Incyte, and Janssen (advisory boards), and an indirect financial relationship (research funding) with Roche, AstraZeneca, Kite/Gilead, Celgene, and MSD. Natacha Bolanos discloses an

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indirect financial relationship with Roche, Ipsen, Kite-Gilead, Janssen, Lilly, Novartis, Sobi, Takeda, and BMS, as the Lymphoma Coalition has received sponsorship from these companies to support specific activities such as educational, advocacy initiatives, CABs, Global Patient Surveys, and Global Summits. Additionally, I am a member of the Patient Global Advisory Board for Ipsen. Martin Dreyling discloses a financial relationship through scientific advisory board honoraria with Abbvie, Astra Zeneca, AvenCell, Beigene, BMS, Genmab, Gilead/Kite, Incyte, Janssen, Lilly/Loxo, Novartis, Roche, Sobi; and research support (institution) Abbvie, Gilead/Kite, Janssen, Lilly, Roche. Umberto Vitolo discloses a direct financial relationship with AbbVie, Genmab, and Gilead (advisory board and lecture honoraria), as well as Incyte and MSD (lecture honoraria). Hervé Tilly discloses a direct financial relationship with Roche and an indirect financial relationship with Roche, AstraZeneca, BMS, and AbbVie. Marie José Kersten disclose an indirect financial relationship with Bristol Myers Squibb, Kite/Gilead, Miltenyi Biotech, Novartis, Roche, Adicet Bio, AbbVie, BeiGene, Galapagos, Mustang Bio, and Janssen. Anne-Ségolène Cottereau, Andreas Rosenwald, Ben (Gerben) Zwezerijnen, Maja Marković, Jean-Philippe Jais, Florence Broussais have no conflicts of interest.

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The data that support the findings of this study are openly available in PubMed at https://pubmed.ncbi.nlm.nih.gov.

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APPENDIX

TABLE A1 Levels of evidence and grades of recommendation (adapted from the Infectious Diseases Society of America-United States Public Health Service Grading System).[164]

Levels of evidence	
I	Evidence from at least one large randomized, controlled trial of good methodological quality (low potential for bias) or meta-analyses of well-conducted randomized trials without heterogeneity
II	Small randomized trials or large randomized trials with a suspicion of bias (lower methodological quality) or meta-analyses of such trials or of trials with demonstrated heterogeneity
III	Prospective cohort studies
IV	Retrospective cohort studies or case–control studies
V	Studies without control group, case reports, and experts' opinions
Grades of recommendation	
A	Strong evidence for efficacy with a substantial clinical benefit, strongly recommended
B	Strong or moderate evidence for efficacy but with a limited clinical benefit, generally recommended
C	Insufficient evidence for efficacy or benefit does not outweigh the risk or the disadvantages (adverse events, costs, etc.), optional
D	Moderate evidence against efficacy or for adverse outcome, generally not recommended
E	Strong evidence against efficacy or for adverse outcome, never recommended

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